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Package [furcontainer](#)

Interface JavaContainer<T>

- public interface JavaContainer<T>
JavaContainer interface that has the Add,Remove,Size,getIterator methods.

- ◦ **Method Summary**

All Methods [Instance Methods](#) [Abstract Methods](#)

Modifier and Type	Method	Description
void	Add(T obj)	
furcontainer.Iterator< T >	getIterator()	
void	Remove(T obj)	
int	Size()	

- ◦ **Method Detail**

- **Add**

void Add([T](#) obj)

- **Remove**

void Remove([T](#) obj)

- **Size**

int Size()

- **getIterator**

furcontainer.Iterator<[T](#)> getIterator()

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Package [furcontainer](#)

Class `JavaVector<T>`

- `java.lang.Object`
- ◦ `furcontainer.JavaVector<T>`

• `public class JavaVector<T>`
 extends `java.lang.Object`

- ◦ **Constructor Summary**

Constructors

Constructor	Description
JavaVector (<code>int vcapacity</code>)	Constructor that takes the capacity to create a vector(array) with that capacity and setting the size to 0.

- **Method Summary**

All Methods [Instance Methods](#) [Concrete Methods](#)

Modifier and Type	Method	Description
<code>void</code>	Add (T obj)	Add() method to add a new element to our vector.
<code>boolean</code>	equals (<code>java.lang.Object obj</code>)	boolean overridden method 'equals' to return true or false if the two vectors are equal or not.
<code>boolean</code>	equals2 (<code>java.lang.Object obj</code>)	boolean method 'equals2(equals to)' to return true or false if the element is in the vector or not.
<code>furcontainer.Iterator<T></code>	getIterator ()	method that will return an iterator using Iterator class.
<code>void</code>	Remove (T obj)	Remove() method to remove a element from our vector.
<code>int</code>	Size ()	Size() method to return the size of the vector.
<code>java.lang.String</code>	toString ()	ToString() method to print the elements of the vector by using our iterator from Iterator class to iterate through the vector and a result String to concetanate the result.

- **Methods inherited from class `java.lang.Object`**

`clone`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

- ◦ **Constructor Detail**

- **JavaVector**

`public JavaVector(int vcapacity)`

Constructor that takes the capacity to create a vector(array) with that capacity and setting the size to 0.

Parameters:
vcapacity -

◦ Method Detail

▪ Add

```
public void Add(T obj)
```

Add() method to add a new element to our vector. It first checks if the element to be added is in the vector or not using equals() method. If the element is already in the vector an exception will be thrown. If the element is not in the vector first capacity will be checked using checkCapacity() method then the element will be added to the end and the size will be incremented.

▪ Remove

```
public void Remove(T obj)
```

Remove() method to remove a element from our vector. It first checks if the element to be removed is in the vector or not using equals() method. If the element to be removed is not in the vector an exception will be thrown. If the element is in the vector then vector will be shifted left after the element to be removed to remove the element. after that the size will be decremented.

▪ Size

```
public int Size()
```

Size() method to return the size of the vector.

▪ **getIterator**

```
public furcontainer.Iterator<T> getIterator()
```

method that will return an iterator using Iterator class.

▪ toString

```
public java.lang.String toString()
```

ToString() method to print the elements of the vector by using our iterator from Iterator class to iterate through the vector and a result String to concetanate the result. result String will be returned after all the merging operations to the result String.

Overrides:

toString in class java.lang.Object

▪ equals

```
public boolean equals(java.lang.Object obj)
```

boolean overridden method 'equals' to return true or false if the two vectors are equal or not. It first checks if obj and our class are the same then it checks sizes are the same then every element of our array(vector) and others is equal then if all of them are equal and ordered it returns true.

Overrides:

equals in class java.lang.Object

▪ equals2

```
public boolean equals2(java.lang.Object obj)
```

boolean method 'equals2(equals to)' to return true or false if the element is in the vector or not. It uses enhanced for loop to iterate through the vector using keyword 'var' to determine its type and compares the current index value with the element to be found in vector.

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Package [furcontainer](#)

Class `JavaSet<T>`

- `java.lang.Object`
 - `furcontainer.JavaSet<T>`

public class `JavaSet<T>`
extends `java.lang.Object`

- **Constructor Summary**

Constructors

Constructor	Description
JavaSet (int jcapacity)	Constructor that takes the capacity to create a set(array) with that capacity and setting the size to 0.

- **Method Summary**

All Methods [Instance Methods](#) [Concrete Methods](#)

Modifier and Type	Method	Description
void	Add (T obj)	Add() method to add a new element to our set.
boolean	equals (<code>java.lang.Object</code> obj)	boolean overridden method 'equals' to return true or false if the two sets are equal or not.
boolean	equals2 (<code>java.lang.Object</code> obj)	boolean method 'equals2(equals to)' to return true or false if the element is in the set or not.
<code>furcontainer.Iterator<T></code>	getIterator ()	method that will return an iterator using Iterator class.
void	Remove (T obj)	
int	Size ()	Size() method to return the size of the set.
<code>java.lang.String</code>	toString ()	ToString() method to print the elements of the set by using our iterator from Iterator class to iterate through the set and a result String to concetanate the result.

- **Methods inherited from class `java.lang.Object`**

`clone`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

- **Constructor Detail**

- **JavaSet**

`public JavaSet(int jcapacity)`

Constructor that takes the capacity to create a set(array) with that capacity and setting the size to 0.

Parameters:
jcapacity -

◦ Method Detail

▪ Add

```
public void Add(T obj)
```

Add() method to add a new element to our set. It first checks if the element to be added is in the set or not using equals() method. If the element is already in the set an exception will be thrown. If the element is not in the set first capacity will be checked using checkCapacity() method then the element will be added to the end and the size will be incremented.

▪ Remove

```
public void Remove(T obj)
```

▪ Size

```
public int Size()
```

Size() method to return the size of the set.

▪ **getIterator**

```
public furcontainer.Iterator<T> getIterator()
```

method that will return an iterator using Iterator class.

▪ toString

```
public java.lang.String toString()
```

ToString() method to print the elements of the set by using our iterator from Iterator class to iterate through the set and a result String to concatenate the result. result String will be returned after all the merging operations to the result String.

Overrides:

toString in class java.lang.Object

▪ equals

```
public boolean equals(java.lang.Object obj)
```

boolean overridden method 'equals' to return true or false if the two sets are equal or not. It first checks if obj and our class are the same then it checks sizes are the same then every element of our array(set) and others is equal then if all of them are equal it returns true. Unless vector elements does not have to be ordered.

Overrides:

equals in class java.lang.Object

▪ equals2

```
public boolean equals2(java.lang.Object obj)
```

boolean method 'equals2(equals to)' to return true or false if the element is in the set or not. It uses enhanced for loop to iterate through the set using keyword 'var' to determine its type and compares the current index value with the element to be found in set.

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Package [furcontainer](#)

Class Iterator<T>

- java.lang.Object
 - furcontainer.Iterator<T>

public class Iterator<T>
extends java.lang.Object

Iterator class that has 'hasNext' and 'next' methods.

- ◦ **Method Summary**

All Methods [Instance Methods](#) [Concrete Methods](#)

Modifier and Type	Method Description
boolean	hasNext() boolean hasNext() method to return true or false if there is another element in the container according to the size value.
java.lang.Object	next() next() method to return the next element in the container and incrementing the index so the iterator.

▪ **Methods inherited from class java.lang.Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

- ◦ **Method Detail**

▪ **hasNext**

public boolean hasNext()

boolean hasNext() method to return true or false if there is another element in the container according to the size value.

▪ **next**

public java.lang.Object next()

next() method to return the next element in the container and incrementing the index so the iterator.

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Package [furcontainer](#)

Class DriverCode

- java.lang.Object
 - furcontainer.DriverCode

public class DriverCode
extends java.lang.Object

Driver class to test the implemented methods and write sets and vectors to text files. It gets an input between 1-3 to either auto test,create a vector or create a set. For 1 it automatically creates a vector and a set and it tries all the methods for the vector and set such as Add,Remove,ToString,Size and equals finally. Finally it writes these set and vector to file using the Iterator created. For 2 and 3 it wants input to take the capacity from the user checking for validity while capacity>1 After that in a menu it wants the user to try the functions Add,Remove,Size,ToString and finally if 5 is chosen It will write the set or vector to file using Iterator and terminate the program. As before it will check validity for the input to be between 1-5.

- ◦ **Constructor Summary**

Constructors

Constructor Description

[DriverCode\(\)](#)

- **Method Summary**

All Methods [Static Methods](#) [Concrete Methods](#)

Modifier and Type	Method	Description
static void	Auto_testContainer (furcontainer.JavaContainer<java.lang.Integer> container)	Auto test method to test the created vector and set as container It tests Add,Remove,Size methods and ToString method according to the class instance.
static void	CreateSet (furcontainer.JavaContainer<java.lang.String> set)	If the choice is 3 to create a set with user input.
static void	CreateVector (furcontainer.JavaContainer<java.lang.String> vector)	If the choice is 2 to create a vector with user input.
static void	main (java.lang.String[] args)	

▪ Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

- - **Constructor Detail**

- **DriverCode**

```
public DriverCode()
```

- **Method Detail**

- **main**

```
public static void main(java.lang.String[] args)
```

- **Auto_testContainer**

```
public static void Auto_testContainer(furcontainer.JavaContainer<java.lang.Integer> container)
```

Auto test method to test the created vector and set as container It tests Add,Remove,Size methods and ToString method according to the class instance. It throws an exception if the element to be added is already in the set(only for set) and element to be removed does not exist in the set or vector.

Parameters:

container -

- **CreateVector**

```
public static void CreateVector(furcontainer.JavaContainer<java.lang.String> vector)
```

If the choice is 2 to create a vector with user input. It creates the vector using the CreateVector method and adds,removes elements and returns the size and displays it according to the user input.

Parameters:

vector -

- **CreateSet**

```
public static void CreateSet(furcontainer.JavaContainer<java.lang.String> set)
```

If the choice is 3 to create a set with user input. It creates the set using the CreateSet method and adds,removes elements and returns the size and displays it according to the user input.

Parameters:

set -

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